

Oxytocin prevents neuronal network pain-related changes on spinal cord dorsal horn *in vitro*

Irma A. Tello-García^a, Jesús Pérez-Ortega^b, Guadalupe Martínez-Lorenzana^a,
Abimael González-Hernández^a, Miguel Condés-Lara^{a,*}¹

^a Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus Juriquilla, Boulevard Juriquilla, No. 3001, C.P. 76230, Querétaro, Mexico

^b The NeuroTechnology Center, Columbia University, Biological Sciences, 901 NWC Building, 550 West 120th Street, New York, N.Y., 10027, United States

ARTICLE INFO

Keywords:

Calcium imaging
Oxytocin
Antinociceptive
Dorsal horn

ABSTRACT

Recently, oxytocin (OT) has been studied as a potential modulator of endogenous analgesia by acting upon pain circuits at the spinal cord and supraspinal levels. Yet the detailed action mechanisms of OT are still undetermined. The present study aimed to evaluate the action of OT in the spinal cord dorsal horn network under nociceptive-like conditions induced by the activation of the *N*-methyl-D-aspartate (NMDA) receptor and formalin injection, using calcium imaging techniques. Results demonstrate that the spontaneous Ca^{2+} -dependent activity of the dorsal horn cells was scarce, and the coactivity of cells was mainly absent. When NMDA was applied, high rates of activity and coactivity occurred in the dorsal horn cells; these rates of high activity mimicked the activity dynamics evoked by a neuropathic pain condition. In addition, although OT treatment increased activity rates, it was also capable of disrupting the conformation of coordinated activity previously consolidated by NMDA treatment, without showing any effect by itself. Altogether, our results suggest that OT globally prevents the formation of coordinated patterns previously generated by nociceptive-like conditions on dorsal horn cells by NMDA application, which supports previous evidence showing that OT represents a potential therapeutic alternative for the treatment of chronic neuropathic pain.

1. Introduction

Recent evidence has described oxytocin (OT) as an inhibitor of nociceptive activity in either the central nervous system [1–5] or peripheral nerve terminals [6]. OT has also been reported as a successful alternative for pain management in humans [7,8]. OT administration demonstrated a striking antinociceptive effect by reducing the pain-related signals and perception. Particularly, evidence shows that OT is capable of diminishing the pain-related electrical activity of spinal cord dorsal horn neurons and therefore inhibiting the transmission of antinociceptive information [1]. However, the intrinsic participation of OT and its effect on inhibiting the transmission of nociceptive information in the functional networks of sensory processing of the dorsal horn circuitry has remained elusive.

On the other hand, most works done on the spinal cord have pursued a better and more profound understanding of the histological

aspects of this region. For several years' electrophysiology has been the standard technique used to study the physiology of the neuronal components of the dorsal horn because of its excellent temporal resolution and sensitivity. Nevertheless, the functional networks of sensory processing were not completely accessible with the main techniques used in the last century. Calcium imaging techniques with epifluorescent or two-photon microscopy allow the observation and analysis of large areas with several neurons. Some studies done in intact [9–16] or *in vitro* spinal cord tissue have been beneficial in understanding how cells function and respond to different stimuli [17,18].

To address the changes produced by OT in the neuronal network circuits of the dorsal horn, we used calcium imaging analysis on spinal cord slices to describe overall neuronal activity in spinal cord dorsal horn slices *in vitro* on a circuit that had not been altered by any stimulus. Afterward, we challenged the system with the application of *N*-methyl-D-aspartate (NMDA) to simulate the nociceptive-related

Abbreviations: OT, oxytocin; NMDA, *N*-methyl-D-aspartate; NMDAr, *N*-methyl-D-aspartate receptors

* Corresponding author at: Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus Juriquilla, Lab B-16. Boulevard Juriquilla, No. 3001, C.P. 76230, Querétaro, Mexico.

E-mail address: condes@unam.mx (M. Condés-Lara).

¹ http://132.248.142.23/web_site/home_pages/35.

<https://doi.org/10.1016/j.ceca.2020.102246>

Received 23 April 2020; Received in revised form 2 June 2020; Accepted 16 June 2020

Available online 18 June 2020

0143-4160/© 2020 Published by Elsevier Ltd.

activation of dorsal horn neurons [17,19]. Finally, we show that OT application inhibits the pain-related changes in neuronal circuit dynamics produced by NMDA.

2. Experimental Procedures

2.1. Experimental animals and ethical standards

A total of 35 both sex neonatal rats of 8–12 days old (P8–P12) were used in the present experiments. The animals were housed in clear plastic cages and maintained on a 12:12 h light–dark cycle (lights on at 07:00 h) at room temperature ($22^{\circ}\text{C} \pm 2^{\circ}\text{C}$), with food and water *ad libitum*. All experiments followed the National Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH publication no. 80-23, revised 1996) and the International Association for the Study of Pain (IASP) Guidelines for the Use of Animals in Research [20], and were conducted with the approval of the Bioethics Committee at the Instituto de Neurobiología (Bioethical protocols numbers 008–010). All efforts were made to limit distress and use only the number of animals necessary to produce reliable scientific data.

2.2. General methods

2.2.1. Slice preparation

Rats were anesthetized with pentobarbital (50 mg/kg) and perfused transcardially with ice-cold ($\sim 4^{\circ}\text{C}$) saline-sucrose solution (sucrose, 238 mM; glucose, 30 mM; NaHCO_3 , 25 mM; KCl, 3 mM; MgCl_2 , 2.5 mM; oxygenated with 95 % O_2 , 5 % CO_2 ; pH = 7.4). Then, the lumbosacral section of the spinal cord was extracted and mounted on an NVSL Vibroslice™ (World Precision Instruments Inc., Sarasota, FL, USA), and 500–550 μm thick transverse slices were obtained. Slices remained for at least 1 h before experiments at room temperature ($20\text{--}24^{\circ}\text{C}$) in saline solution containing (in mM): 119 NaCl, 30 Glucose, 25 NaHCO_3 , 6 KCl, 2.5 MgCl_2 , 1.5 CaCl_2 , oxygenated with 95 % O_2 , 5 % CO_2 , pH = 7.4.

2.2.2. Calcium imaging and treatments

Slices were incubated in dark conditions for 30–60 min at room temperature in a solution with 1 mL of saline plus 1.8 % D-mannitol (Sigma-Aldrich Co., St. Louis, MO, USA; CAS No. 69-65-8) and 20 μL of a solution containing 5 μg of Fluo-8® AM (AAT Bioquest Inc., Sunnyvale, CA, USA; CAS No. 1345980-40-6) diluted in 50 μL dimethylsulphoxide (DMSO), and equilibrated with 95 % O_2 and 5 % CO_2 . To obtain the Ca^{2+} fluorescence signals, a motorized inverted Olympus IX81 fluorescence microscope equipped with a UPlanSApo 10x/0.40na objective was used (Olympus Latin America, Inc, Florida, FL, USA). To record the Ca^{2+} fluorescence transient signals, light pulses at 488 nm (35–100 msec exposure) were delivered to the spinal cord slice using a xenon lamp (cell-R DSU module, Olympus) connected to the microscope with an optical fiber. Changes in Ca^{2+} fluorescence signals were recorded with an ORCA®-R² digital CCD camera (Hamamatsu Photonics K.K. Japan) that gives an image field of $866 \times 660 \mu\text{m}$ (see Fig. 1A, B). Short videos ($\sim 240 \text{ s} = \text{epoch}$; 4–5 fps) were acquired with the xCellence-RT imaging software, v1.2 (Olympus Soft Imaging Solution GmbH, Germany) at 200–250 msec time intervals.

At the end of the experiment, potassium chloride solution (300 μL of KCl at 500 mM in 3 mL of the recording chamber solution, giving a concentration of 50 mM) was applied to the recording chamber as a positive control. Neurons active (15–124 depending on each experimental condition) during the experiment were identified due to their Ca^{2+} transients and recorded in the following groups:

- Control group: no treatment was added to the slice. We made recordings of basal activity of dorsal horn cells, in order to observe how the spontaneous Ca^{2+} activity is at the dorsal horn.
- NMDA group: 8 μM NMDA [24–26] was added to the sliced preparation and preincubated for 5 min before recordings to evaluate the possible changes in the calcium-dependent activity of spinal dorsal horn cells under excitatory input.
- NMDA + OT group: 8 μM NMDA + 5 μM OT. OT was added after NMDA preincubation in order to evaluate if OT would have an effect on the activity first promoted by NMDA pretreatment. Experiments for NMDA + OT were preincubated for 5 min with NMDA, after which OT was added. Five minutes later, the recordings were carried out (10 min total).
- OT group: 5 μM OT [4–6,35,40] was added to the slice in order to evaluate the effect of OT on spontaneous activity. OT experiments were preincubated 5 min before recordings; no NMDA was added to the preparation.
- Formalin 5% group: injection of formalin 5% was done 48 h before calcium imaging recordings, in order to evaluate the calcium-dependent activity of dorsal horn cells under a neuropathic pain-like condition and compare if the application of NMDA could reproduce this activity pattern as a neuropathic pain model for calcium imaging of spinal cord slices.

2.3. Image analysis

Initial image processing was carried out using Im-Patch© (<http://impatch.ifc.unam.mx/>), an open-source software to analyze changes in fluorescence Ca^{2+} signals. This program was written in LabVIEW programming environment (National Instruments™, USA) by one of the co-authors of this paper and has been extensively validated in several reports analyzing Ca^{2+} fluorescent signals [21–23]. In brief, this program allowed us to extract a summary image, which was obtained from the standard deviation from each video; then, localization coordinates of neurons active during the experiment were automatically detected by pattern recognition using circular templates with a 3-pixel diameter ($\sim 4 \mu\text{m}$) (Fig. 1C). Calcium transients, represented as changes in fluorescence ($F_i - F_0$)/ F_0 (where F_i is the fluorescence intensity and F_0 the basal fluorescence of each neuron), from each region of interest were extracted over time (Fig. 1D). Photobleaching, noise, and airy disk phenomena were corrected as previously described [23–25].

2.3.1. Detection of neuronal activity

It has been reported that the first-time derivative of the Ca^{2+} transient reflects the electrical discharge of neurons; specifically, the positive time derivative of a Ca^{2+} transient is similar to the duration of the firing trains in neurons [23–25]. Therefore, frames with active neurons (seen with its derivative) were taken as an indirect measure of the electrical activity of neurons over time (global activity); in all cases, only amplitudes 2.5 times above the standard deviation of the noise value were considered for analysis. All recordings were examined manually to remove Ca^{2+} transients attributed to glial cells, which have been described as slower than neuron Ca^{2+} signals (minutes long) [23–25]. Detection of neuronal activity was driven as previously described [23–25]. Briefly, a binary matrix ($N \times F$) that represents the activity of neurons recorded simultaneously over time was constructed with the data (one matrix per epoch), where 1 represents an active and 0 an inactive neuron (plotted as Fig. 1E). This binary matrix was called experimental matrix, where every row corresponds to an active neuron (N), while columns represent the number of frames (F) on the video, as follows: 1 column is equal to 1 frame. The sum of every column is a coactivity vector ($1 \times F$) that represents the coactivity of several neurons over time (plotted as Fig. 1F).

2.3.2. Coactivity analysis

After the experimental matrix was constructed, coactivity analysis was performed on a custom-made program, written in MATLAB (MathWorks, Inc. MA, USA) by a co-author of this paper, that also includes some algorithms from the Brain Connectivity Toolbox (<https://>

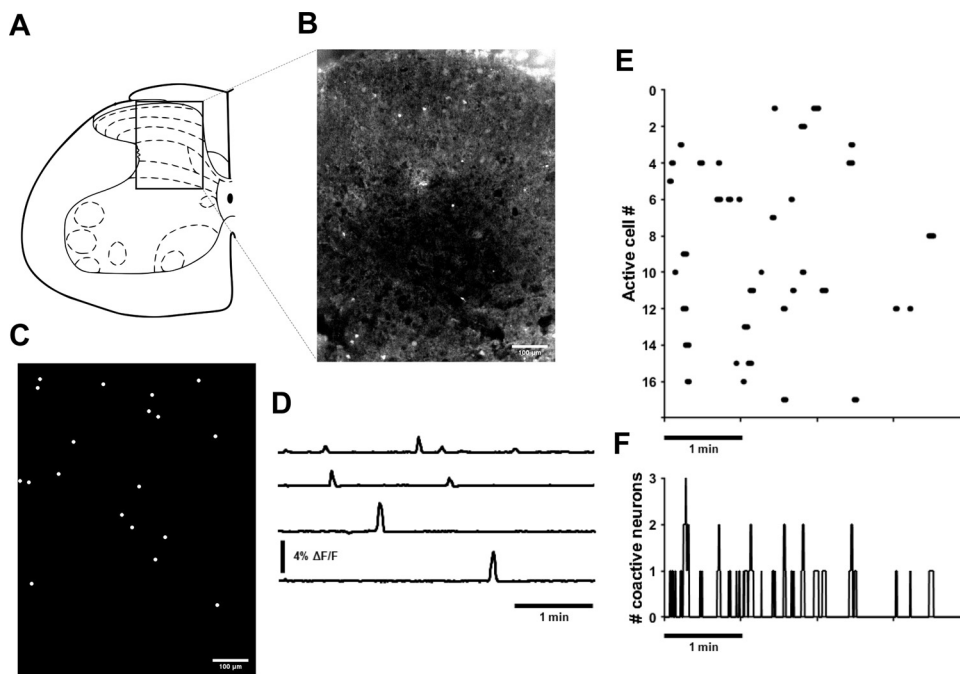


Fig. 1. Calcium imaging in spinal cord slice. (A) Scheme of the spinal cord area where calcium imaging was performed. the area corresponds to Rexed dorsal horn laminae I-V ($866 \times 660 \mu\text{m}$). (B) Image of spinal cord tissue area recorded for 4 min. The spinal cord slice was loaded with Fluo-8 AM; spontaneous activity was recorded (scale bar = $100 \mu\text{m}$). (C) Automatic selection of active neurons and their coordinates on the tissue area recorded during the experiment. (D) Examples of transient increases of calcium of 4 neurons during recordings ($\Delta F/F$). (E) Raster plot showing the spontaneous activity of > 15 neurons in the dorsal horn of the spinal cord. Each row represents the activity of a single neuron across the series of images; cells are ordered and numbered from superficial to deep one (top to bottom, y-axis), and columns are movie frames converted to minute scale (x-axis, scale bar = 1 min). (F) The activity histogram shows the summed neuronal activity from the raster plot frame by frame and represents the number of coactive neurons along with the recordings (scale bar = 1 min).

sites.google.com/site/bctnet/). In this program, coactivity vectors from every experimental matrix were obtained, and the probability that a given peak appeared by chance was assessed by determining a significance threshold using Monte Carlo permutations (1000 times) as previously described [23–25]. The significance threshold was established as $p < 0.05$, and is marked as a dashed line on every coactivity histogram [23–25]; also, all the peaks that surpassed the significance level are denoted in red (Fig. 3B). For analysis, significant peaks of neuronal coactivity were stored in a matrix of coactivity peaks ($A \times P$), where A is the number of active cells in every peak, and P is the number of coactivity peaks. The Jaccard similarity coefficient was measured to evaluate if coactivity vectors were different from one another [23–25]. Additionally, to evaluate an increase or decrease in synchronization of activity, coactivity vectors were counted in each epoch.

2.4. Drugs

Apart from the anesthetic (pentobarbital sodium; PiSA®, Mexico), and other reagents mentioned above, this study also used the following compounds (obtained from the sources indicated): *N*-methyl-D-aspartic acid (NMDA; CAS No. 6384-92-5) and oxytocin acetate salt hydrate (OT; CAS No. 50-56-6) purchased from Sigma-Aldrich Co. (St. Louis, MO, USA). OT and NMDA stock solutions were prepared before experiments and added to the recording chamber at the final concentration of $5 \mu\text{M}$ OT and $8 \mu\text{M}$ NMDA. For some experiments, formalin 5% ($15 \mu\text{L}$, diluted in saline) was injected subcutaneously (s.c.) into the dorsal surface of the hind paw 48 h prior to spinal cord extraction. NMDA and OT were dissolved in isotonic saline solution (0.9 % NaCl).

2.5. Data and statistical analysis

Most statistical methods used have been previously described [23–25]. Briefly, the number of cells was analyzed and the total activity during the epoch and number of coactivity peaks were considered for each experimental acute *in vitro* condition and plotted as the mean $\pm$ standard error of the mean (SEM). A non-matched Kruskal-Wallis with Dunn's *post hoc* test was used to analyze the differences between each group compared to the control condition and within groups.

We plotted the total activity against the coactivity peaks to compare the relation between both variables and performed a linear regression

analysis to obtain the slope values and significance of the correlation of the variables. In addition, the global activity of every condition over time was represented by cumulative activity distributions, and the rates of accumulation were approximated with *ad hoc* linear regressions [23–26]. We considered statistical difference values as $p < 0.05$.

3. Results

We recorded an area of $866 \times 660 \mu\text{m}$ of the spinal dorsal horn. This area covers the Rexed lamina I to V of the spinal dorsal horn (see Fig. 1A and a representative image of the tissue from the slice analyzed in Fig. 1B). The video recordings of the area visualized were further analyzed as previously described in the *Image analysis* section, and a summary image was obtained; from this summary image, an automatic localization of active neurons was performed. The coordinates of each neuron were used to identify every cell in the slice analyzed, as shown in Fig. 1C. Fig. 1D exemplifies representative Ca^{2+} transients from four spontaneous active neurons of the dorsal horn during 4 min of recording (epoch). Every epoch was plotted in a raster plot to visualize activity from several active neurons, where dots represent neuronal activity, the number of every active cell on the y-axis (organized from superficial to deeper lamina, I to V, in ascending order), and the temporal line of the experiment on the x-axis (Fig. 1E). An activity histogram is shown below every raster plot, and this represents the coactivity of the neurons over time; in this case, each peak is a coactivity vector extracted from the raster plot (Fig. 1F). In general terms, as validated in striatal neuronal tissues [23–26], the sequence of coactivity peaks could denote the activity sequence of different groups of several neurons alternating their activity during the recordings. In this sense, to analyze the role of OT in the spinal neuronal pain network, we analyzed the Ca^{2+} transients under different conditions.

3.1. NMDA increases the Ca^{2+} -dependent activity of cells in the dorsal horn circuit

First, we characterize the spontaneous Ca^{2+} -dependent activity in dorsal horn cells in Control situation (Fig. 2A). Our results revealed a scarce Ca^{2+} activity in Control situation. In Fig. 2B and C, the Ca^{2+} dependent activity in Control situation shows that the number of cells involved was low (28.71 ± 5.56 , $n = 7$ slices) and the activity of these

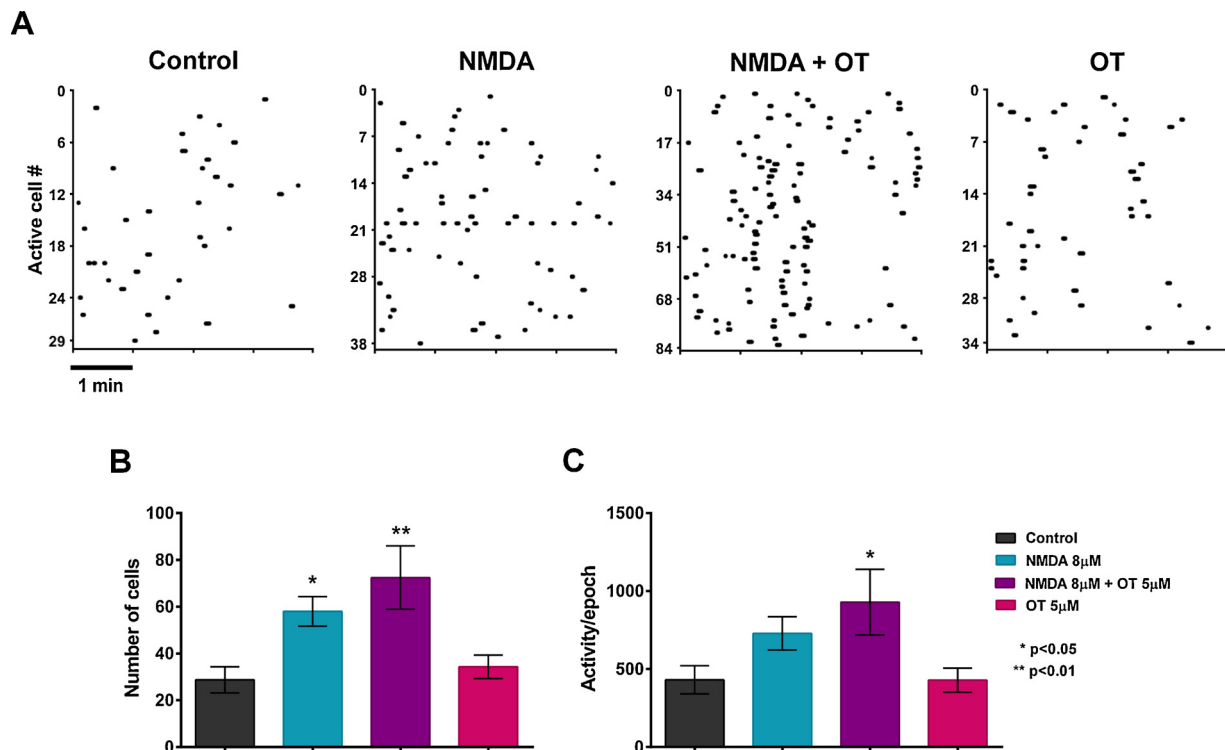


Fig. 2. NMDA increases the activity of spinal cord dorsal horn neurons.

(A) Representative raster plot of the different groups: control, 8 μM NMDA, 8 μM NMDA + 5 μM OT, and 5 μM OT; every epoch shows 4 min of activity of dorsal horn neurons. Note scarce control activity and an increase under NMDA and NMDA + OT. (B) Active cell number. The graph compares the number of active cells during epoch for each group; clearly, NMDA and NMDA + OT treatments increase the number of active cells. (C) Global activity. The graph compares the total activity for epoch under different treatments. NMDA increases the number of active cells and partially increases global activity compared with the control condition. In addition, NMDA + OT co-application increased significantly not only the number of cells but also the global activity compared with the control condition, indicating that OT recruits extracellular elements in addition to those recruited by NMDA by itself. Data are represented as mean ± SEM, $n = 7$, $p < 0.05$ (*), or $p < 0.01$ (**) compared to control, Kruskal-Wallis with Dunn's *post hoc* test. For treatments, NMDA experiments were preincubated for 5 min; NMDA + OT experiments were preincubated for 5 min with NMDA and then OT was added for another 5 min of preincubation (10 min total); finally, OT experiments were preincubated for 5 min; no NMDA was added to the preparation.

cells (*i.e.*, frames with active neurons) was scarce (431.30 ± 89.68 , $n = 7$ slices) during the 4-minute recordings. This scarcity of spontaneous Ca^{2+} -dependent activity in dorsal horn cells agreed with previous reports [17,18].

Since we were interested in analyzing the effect of OT under an excitatory input, we used NMDA as an excitatory driver to increase the activity of the dorsal horn circuit; certainly, NMDA-sensitive receptors are known as the primary mediators of changes in neuronal activity in spinal Ca^{2+} mobilization related to nociceptive stimuli [17,18]. Similar approach has been used and validated in the spinal cord [24–26].

Under these considerations, we evaluated the action of NMDA application in spinal cord slices upon the Ca^{2+} transient events. Results showed that application of NMDA (8 μM) in spinal cord slices significantly increases the number of active cells (Fig. 2B, 58.00 ± 6.32 , $n = 7$ slices, $p < 0.05$, Kruskal-Wallis, Dunn's post-test) and produces a partial increase in the global activity without being statistically significant when compared with the control group (Fig. 2C, 729.00 ± 106.80 , $n = 7$ slices).

Considering that OT at spinal cord levels could act as an inhibitor of nociceptive activity in animals [4–6,35,36,40] and humans [7,8], we applied NMDA (8 μM) + OT (5 μM) to evaluate if OT modifies the Ca^{2+} activity evoked by NMDA (as nociceptive-like stimuli). For treatment co-application details, see section 2.2.2 Calcium imaging and treatments. We found that after the co-application of NMDA + OT, the number of active cells increases similarly to the NMDA group, when compared with the control group (Fig. 2B, 72.43 ± 13.56 , $n = 7$ slices, $p < 0.01$, Kruskal-Wallis, Dunn's post-test). Besides, the co-application of NMDA + OT increases not only the number of active cells but also

the overall activity (Fig. 2C, 928.70 ± 210.40 , $n = 7$ slices, Kruskal-Wallis, Dunn's post-test) when compared with the control group.

Finally, the analysis of OT (5 μM) effects shows that the number of active cells and the global activity were similar to those of the control group (Fig. 2B, C, 34.29 ± 5.02 active cells, 428.70 ± 77.49 global activity, $n = 7$ slices). In brief, control Ca^{2+} activity and the cells number was scarce, NMDA increased the Ca^{2+} activity and the active cells number, NMDA + OT presented more Ca^{2+} activity and cells number participating and finally, OT along present an Ca^{2+} activity and cells number like in control situation.

3.2. Oxytocin prevents neuronal network dynamic changes produced by NMDA in dorsal horn cells

Afterward, we characterize if the increase in the number of active cells induced by NMDA produces changes in the activity dynamics of the dorsal horn circuit. Based on similar studies from striatal structures, where NMDA produces changes in activity dynamics of cells [23–25], we analyzed the coactivation of cells in the spinal cord as previously described [25].

Again, raster plots show the activity of dorsal horn cells over time under different conditions (Fig. 3A); colored dots represent the activity of neurons with statistical significance for coactivity. Coactivity histograms, below raster plots, represent the summed neural activity of every epoch (Fig. 3B). Every coactivity peak represents a group of active cells at the same time. After Monte Carlo permutations (see 2.3.2 Coactivity analysis), all the synchronization peaks denoted in red surpassed the significance threshold represented in coactivity histograms

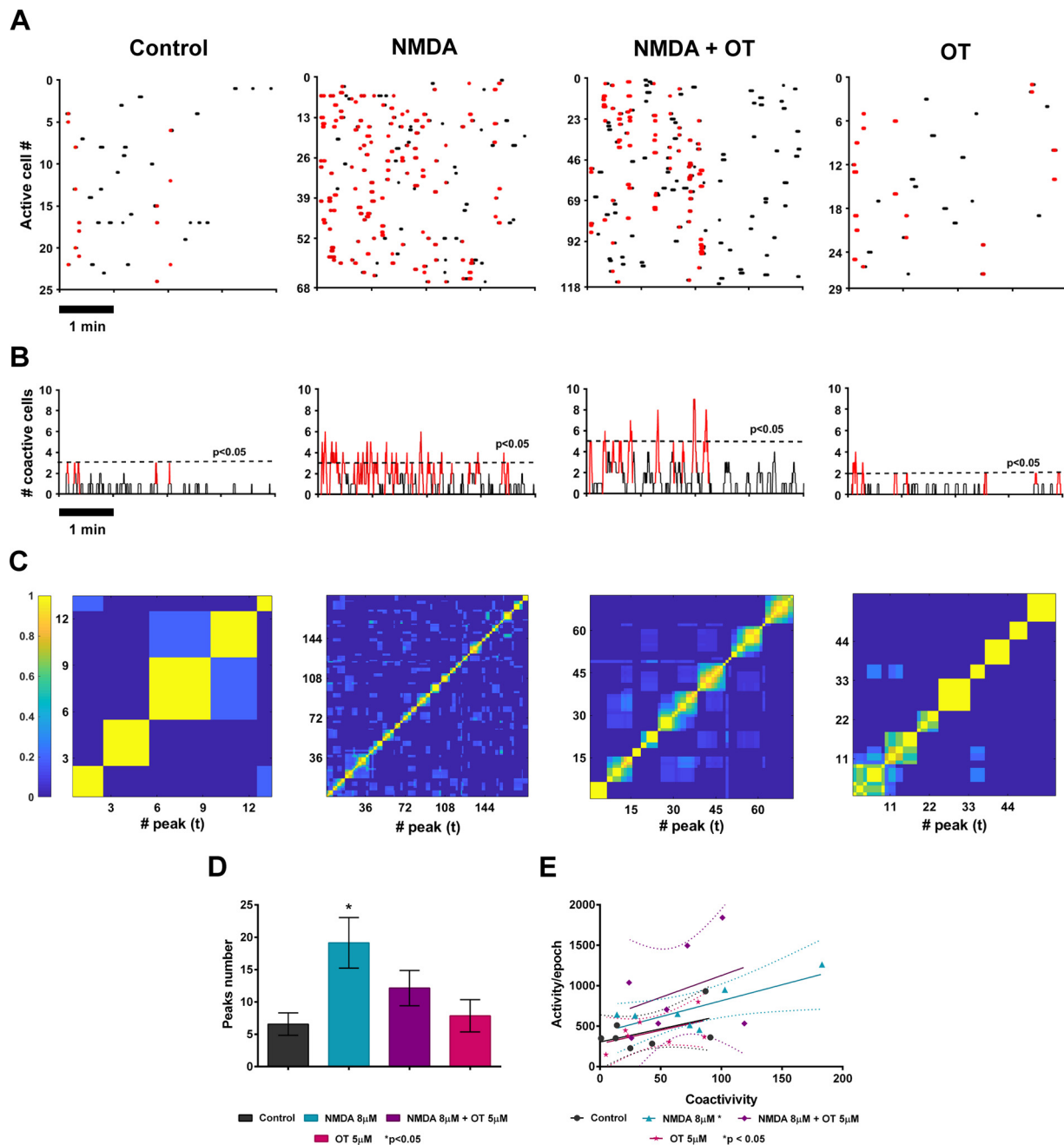


Fig. 3. Oxytocin treatment reduces the coordination of neuronal network circuits produced by NMDA in the spinal cord dorsal horn.

(A) Representative raster plots of the different groups; red dots represent the activity of sets of neurons with simultaneous recurrent activity, $p < 0.05$. (B) Coactivity histograms represent the number of coactive cells for each condition; peaks of synchronization denoted in red surpassed the significance threshold (represented as a dashed horizontal line). (C) Cross-correlation matrix of similarity index; compares all possible coactivity vectors paired over time for every treatment. (D) Number of coactivation peaks; the graph represents the appearance of synchronization peaks for each group. Note that OT treatment reduces the number of coactivity peaks produced by NMDA application to levels similar to those of the control condition. (E) Correlation between the global activity of epochs (Y-axis) vs. coactivity (X-axis) for each condition. When comparing the number of active frames (global activity) versus the relative coactivity, linear regression analysis indicated that only NMDA groups showed a correlation between the two variables, $p < 0.05$. Results suggest that OT treatment reduces the coordination of the activity produced by NMDA. The color scale bar in C represents the correlation between the coactivity vectors presented on every representative correlation matrix. Data are presented as mean $\pm$ SEM in D, solid lines represent the linear regression, and dotted lines represent the 95 % of the confidence interval, $n = 7$, $p < 0.05$ (*) compared to control in D (Kruskal-Wallis with Dunn's *post hoc* test) or different than slope zero in E.

as a dashed horizontal line (Fig. 3B). We found few significant spontaneous peaks of coactivity in the control group (Fig. 3D, 6.57 ± 1.73 , $n = 7$ slices). In contrast, NMDA (8 μ M) application promotes changes in the activity dynamics of spinal cord circuitry characterized by an increase in the number of coactive cells, counted as the number of coactivity peaks (Fig. 3D, 19.14 ± 3.91 , $n = 7$, $p < 0.05$, Kruskal-Wallis, Dunn's *post-test*), indicating that pools of neurons synchronize

their activity and alternate with other pools of neurons over time. Interestingly, the co-application of NMDA (8 μ M) + OT (5 μ M) prevents changes in coactivity dynamics, reducing the number of coactive peaks to non-statistically significant levels when compared with the control group (Fig. 3D, 12.14 ± 2.72 , $n = 7$). Also, OT (5 μ M) application does not produce any significant change in the coactive peak number compared to the control group (Fig. 3D, 7.85 ± 2.49 , $n = 7$).

We also measured the Jaccard similarity index of all significant coactivity vectors, and we compared whether two vectors have the same components (neurons) or not from all possible pairs of vectors. Then, we plotted as a cross-correlation matrix in a pseudo-color scale (Fig. 3C); in this representation, a patchy appearance denotes that similar and several pools of neurons were in charge of activity synchronization over time [23–26]. As seen in Fig. 3C, NMDA produces a high level of synchronization marked as a patchy appearance on the plot.

To evaluate whether the global activity and coactivity of cells were correlated or not, we performed a linear regression analysis comparing the total activity of each experiment with the coactivity peak number. We found that only the NMDA group showed a significant correlation between global activity and coactivity that is not present under any other condition (Fig. 3E, $p < 0.05$, $n = 7$ slices, linear regression). These results suggest that NMDA induces synchronization in the Ca^{2+} -dependent activity of spinal dorsal horn neurons and that OT acts by disrupting this level of synchronization.

3.3. *In vivo* formalin injection emulates increasing activity produced by NMDA in spinal cord slices *in vitro*

Evidence shows that hind paw formalin injection in rats produces long-term pain-related changes in the spinal cord dorsal horn [17,18]. To evaluate if changes in neuronal activity dynamics produced by NMDA *in vitro* application are similar to changes produced by a central sensitization developed by an inflammatory chronic pain model, we injected formalin 5% into the hind paw of the rats, and 48 h later we evaluated if the injection produced *ex vivo* changes in spinal cord neuronal activity. An example of activity evoked by formalin 5% injection is represented as a raster plot in Fig. 4A. By summing the activity over time, a graph of cumulative activity was obtained (Fig. 4B); this graph shows total global activity under different conditions over time (Ad hoc fitting of straight lines, where slopes become the rate of accumulation over time) [14]. An average of experiments of the control group shows 103.50 ± 0.31 (act/min), which increases to 189.07 ± 1.07 (act/min) for the NMDA group and 177.62 ± 0.69 (act/min) for the formalin group. OT shows a rate of accumulation of 113.925 ± 0.362 (act/min) that is similar to the control group, suggesting that OT does not have an effect on the calcium-dependent activity of the intact spinal cord dorsal horn cells along time. Finally, we did not find any statistical differences when comparing NMDA and formalin groups in i) number of cells (54.00 ± 8.28 , $n = 7$ slices, Kruskal-Wallis, Dunn's post-test, data not shown), ii) global activity (shown as cumulative activity on Fig. 4B), and iii) number of coactivity peaks (8.7 ± 2.35 , $n = 7$ slices, Kruskal-Wallis, Dunn's post-test, data not shown). Therefore, results exhibited that the injection of formalin 5% increases Ca^{2+} -dependent activity in dorsal horn cells in a similar way as the NMDA group."

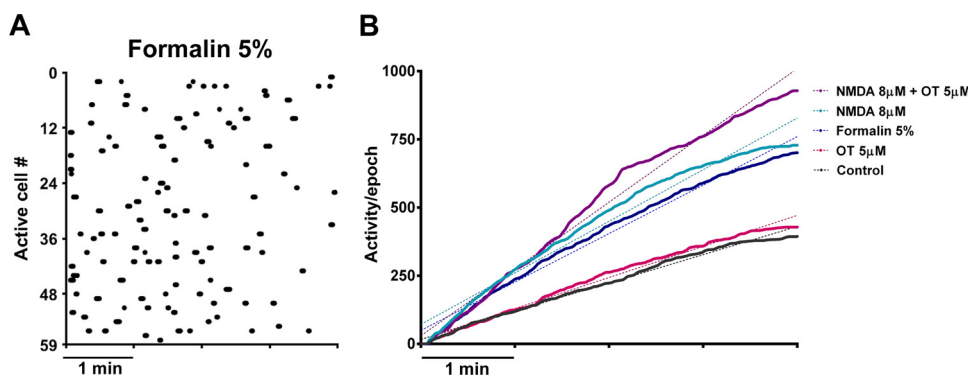


Fig. 4. NMDA activates neuronal network dynamics of the spinal cord dorsal horn similarly to a neuropathic pain condition.

(A) Representative raster plot shows the activity of dorsal horn neurons under formalin 5% injection and (B) Cumulative activity of different conditions during 4-minute recordings. Note that the formalin *in vivo* injection 48 h prior to the *ex vivo* calcium imaging experiment produces changes in the neuronal network dynamics of spinal cord dorsal horn neurons that are mimicked by NMDA application.

4. Discussion and conclusions

Taken together, these results showed the following: (i) The silent nature of the dorsal horn circuit under control conditions. (ii) The dorsal horn circuit can be activated with NMDA application, which increases the number of active cells and the global activity of the circuit; also, NMDA treatment enhances the rates of coordinated activity of the dorsal horn network, regarding neuropathic pain plasticity evoked by the activation of NMDA receptors (NMDAR). (iii) OT can disrupt the coordinated activity patterns of ensembles of neurons in the dorsal horn of the spinal cord, and (iv) formalin injection produces changes in calcium-dependent activity similar to those evoked by NMDA treatment.

In normal conditions, dorsal horn circuitry has been described as silent, both in spinal cord slices [17,18] and in *in vivo* preparations [14]. This correlates with our results that although spontaneous calcium transients occur in dorsal horn neurons under control conditions, scarce synchronous activity is present but only in a few neurons (Figs. 1B and 2D).

Because of the silent nature of the spinal dorsal horn circuit, we used NMDA as an excitatory drive of neuronal activity, in accordance with previous reports showing that NMDA application can induce an increase of calcium-dependent activity in silent networks as striatal structures [23–25]. Additionally, it has been largely documented that NMDAR activation is a crucial mediator of the amplification of nociceptive-related signaling in the spinal cord leading to plasticity mechanisms like central sensitization, which is the major component of inflammatory and neuropathic pain [19,30–33]. Also, activation of the glutamatergic system, and particularly activation of NMDAR, has been implicated in mechanisms of nociceptive related signaling facilitation in the spinal cord like allodynia and hyperalgesia.

We report that NMDA application in spinal cord slices can promote the activity of a silent circuit by increasing the overall calcium-dependent activity and the number of active cells. This is the first study to report the effects of NMDAR activation in a large number of cells (by slice 38–84) covering laminae I–V of the dorsal horn (Fig. 2). Also, our findings correlate with complementary evidence describing that nociceptive peripheral stimuli and electrical stimulation of dorsal roots increase the number and amplitude of calcium transients of neurons on superficial lamina (I–II) of the dorsal horn and that these changes are mediated by the activation of NMDAR [14,16,17,33].

Besides, NMDA also demonstrated to produce changes in the calcium-dependent activity dynamics of the spinal dorsal horn by enhancing the coordinated activity of neurons (Fig. 3). Consistent with this finding, previous data indicated that NMDA application in cultured spinal cord cells could generate coordinated activity that lasting for up to several hours [30]. Also, when imaging the neuronal network activity in the spinal cord during a nociceptive-like *in vivo* condition, the results show evidence of coordinated activity in the spinal cord (for a review see [34]). This synchronous calcium-dependent activity mediated by NMDAR activation was previously reported in spinal cord slices,

specifically in the superficial laminae (I–III) of the dorsal horn, but only under the condition of a synaptic inhibition removal by blocking GABA_A and glycine receptors [18]. Here, we demonstrated that the application of NMDA alone could evoke high rates of coordinated activity on assemblies of dorsal horn neurons (laminae I–V) even when synaptic inhibition remained intact.

To elucidate whether the facilitation of coordinated activity produced by NMDA activation on dorsal horn neurons could be modified by OT treatment, we applied OT to the slices previously activated with NMDA. We found that the application of OT does not reduce the overall activity produced by NMDA on the contrary partially increases the activity (Fig. 2). However, OT is capable of reducing the coordinated activity associated with nociceptive-related calcium signaling (Fig. 3). Evidence indicates that OT can modulate the glutamatergic excitatory response in the spinal cord by reducing the increased firing rate produced by nociceptive-related signaling [1,2,35,36]. Although the reduction of NMDA activity by OT has been described in neuronal cultures [37], evidence shows that the primary source of inhibitory action by OT on the spinal cord dorsal horn neuronal network is due to the overall increase in GABAergic transmission by direct and indirect means [38–40]. First, data indicates that OT directly increases the basal levels of extracellular GABAergic transmission [3,41,42]. Second, OT simultaneously decreases the glutamatergic activity by regulating the expression of the NR1 subunit of glutamatergic receptors and the levels of glutamate in the extracellular space by regulating glutamatergic transporter 1 [42]. Besides, indirect inhibition of the spinal cord dorsal horn circuit by OT action arises from the activation of a subpopulation of glutamatergic neurons that promotes local inhibition by recruiting GABAergic interneurons [43]. However, what we aimed to test in the present work by NMDA application is not an acute pain signaling but a pathological condition at the spinal cord as neuropathic pain condition. Taking that into account, we applied NMDA in order to mimics the phasic tone of NMDA that leads to the consolidation of central sensitization at the spinal dorsal horn that occurs under neuropathic pain-like conditions.

Interestingly, evidence suggested that the inhibitory effect of OT on the spinal cord dorsal horn is effective only upon previous activation of the glutamatergic nociceptive-related signaling. For example, OT counteracts the enhancement of excitatory activity characterized as an increase in frequency and amplitude of excitatory postsynaptic currents produced by activation of TRPV1 by capsaicin in a pain-related model without showing any effect by itself on the intact system [44]. And not only at spinal level but also at peripheral level, where evidence suggests that OT interacts directly or indirectly with TRPV1 receptor and attenuates capsaicin-induced nociception via TRPV1 desensitization without affecting intact cells [45]. This evidence correlates with our results, which showed that OT treatment reduces the coactivation of neuronal activity after NMDA application and that OT applied to control condition slices does not have any significant effects (Fig. 3).

One limitation of the previous evidence regarding OT inhibitory action on nociceptive-related models is the use of single-cell approaches, due to the effects that OT can have on different cell populations depending on the context, denoting the importance of studying the global actions of OT in the microcircuit as a whole. Our data complements these single-cell data measuring the spinal cord as a complete circuit, showing that despite the increase in global calcium-dependent activity and the recruitment of active cells, OT leads to a disruption of the coordination of neuronal network activity, therefore inhibiting nociceptive-related calcium signaling (Fig. 3), promoting an antinociceptive action.

In order to elucidate if this NMDA activation in the spinal cord dorsal horn neurons is related to nociceptive-like signaling, we injected formalin 5% subcutaneously into the dorsal surface of the hind paw 48 h before calcium imaging recordings. We found that formalin injection produces long-lasting changes in calcium mobilization in dorsal horn neurons; these changes were similar to those observed in slices

pre-treated with NMDA (Fig. 4). Our results correlate with evidence showing that, after 24 h, peripheral injury can trigger long-lasting changes in calcium signaling patterns in the dorsal horn neurons [17], therefore supporting the idea that the coordinated activity produced by NMDA activation enhances nociceptive-related changes in the activity dynamics of the spinal cord dorsal horn, similar to a neuropathic pain model. It has been studied that formalin 5% injection on the hind paw of the rat can produce i) long-lasting secondary thermal and mechanical allodynia and hyperalgesia enhanced for 1–3 days after injection and lasted 3–4 weeks [27,28]. ii) hypersensitivity as well as molecular changes at dorsal root ganglion and spinal dorsal horn that resemble those induced by nerve injury, reaching its maximal effects between 1–3 days after injection, these effects remained for at least 14 days [28]. Based on this evidence we used Formalin 5% to induce neuropathic pain-like changes on the activity of the spinal dorsal horn cells. As the maximal effects were seen between 1–3 day after injection in previous reports [27–29], we tested calcium-dependent activity at 24 h and also at 48 h after formalin injection and changes in calcium-dependent activity were mainly observed after 48 h of formalin injection.

To summarize, the results presented here, combined with previous evidence from our group and others, indicate that OT application could represent a potential treatment to prevent the long-lasting changes in calcium mobilization in the dorsal horn regarding central sensitization under neuropathic pain conditions. The dorsal horn of the spinal cord is the site of the first synapse in the pain pathway and is crucial to study the plasticity mechanisms leading to chronic pain. Studying the global changes in network activity dynamics underlying central sensitization and how OT treatment counteracts these changes will prove useful to understand the development and possible treatment of chronic pain.

CRediT authorship contribution statement

Irma A. Tello-García: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Visualization, Writing - original draft. **Jesús Pérez-Ortega:** Software, Validation. **Guadalupe Martínez-Lorenzana:** Methodology, Resources, Supervision. **Abimael González-Hernández:** Supervision, Writing - review & editing. **Miguel Condés-Lara:** Conceptualization, Methodology, Supervision, Resources, Funding acquisition, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare no conflicts of interest.

Acknowledgments

Irma A. Tello-García is a doctoral student from the Programa de Doctorado en Ciencias Biomédicas, Universidad Nacional Autónoma de México (UNAM) and has received CONACyT fellowship (no. 582491). This study was supported by grant no. IN200415 from PAPIIT-UNAM to MCL. The authors thank to Dr. Michael C. Jeziorski and Dr. Arturo G. Isla for their aid during the drafting of this article and to Jessica González Norris for reviewing the grammatical aspects of this paper.

References

- [1] Y. Miranda-Cardenas, G. Rojas-Piloni, G. Martínez-Lorenzana, J. Rodríguez-Jiménez, M. López-Hidalgo, M.J. Freund-Mercier, M. Condés-Lara, Oxytocin and electrical stimulation of the paraventricular hypothalamic nucleus produce antinociceptive effects that are reversed by an oxytocin antagonist, *Pain* 122 (2006) 182–189, <https://doi.org/10.1016/j.pain.2006.01.029>.
- [2] D. A. Robinson, F. Wei, G.D. Wang, P. Li, S.J. Kim, S.K. Vogt, L.J. Muglia, M. Zhuo, Oxytocin mediates stress-induced analgesia in adult mice, *J. Physiol.* 540 (2002) 593–606, <https://doi.org/10.1113/jphysiol.2001.013492>.
- [3] P. Juif, P. Poiseau, Neurohormonal effects of oxytocin and vasopressin receptor agonists on spinal pain processing in male rats, *Pain* 154 (2013) 1449–1456, <https://doi.org/10.1016/j.pain.2013.05.003>.
- [4] A. González-Hernández, A. Espinosa De Los Monteros-Zuñiga, G. Martínez-

- Lorenzana, M. Condés-Lara, Recurrent antinociception induced by intrathecal or peripheral oxytocin in a neuropathic pain rat model, *Exp. Brain Res.* 237 (2019) 2995–3010, <https://doi.org/10.1007/s00221-019-05651-7>.
- [5] E. García-Boll, G. Martínez-Lorenzana, M. Condés-Lara, A. González-Hernández, Oxytocin inhibits the rat medullary dorsal horn Sp5c/C1 nociceptive transmission through OT but not V1A receptors, *Neuropharmacology* 129 (2018) 109–117, <https://doi.org/10.1016/j.neuropharm.2017.11.031>.
 - [6] A. González-Hernández, A. Manzano-García, G. Martínez-Lorenzana, I.A. Tello-García, M. Carranza, C. Arámburo, M. Condés-Lara, Peripheral oxytocin receptors inhibit the nociceptive input signal to spinal dorsal horn wide-dynamic-range neurons, *Pain* 158 (2017) 2117–2128, <https://doi.org/10.1097/j.pain.0000000000001024>.
 - [7] H. Zayas-González, A. González-Hernández, A. Manzano-García, D. Hernández-Rivero, M.A. García-Cuevas, J.C. Granados-Mortera, L. Vaca-Aguirre, S. Flores-Fierro, G. Martínez-Lorenzana, M. Condés-Lara, Effect of local infiltration with oxytocin on hemodynamic response to surgical incision and postoperative pain in patients having open laparoscopic surgery under general anesthesia, *Eur. J. Pain* 23 (2019) 1519–1526, <https://doi.org/10.1002/ejp.1427>.
 - [8] M. Condés-Lara, H. Zayas-González, A. Manzano-García, E. Córdova-Quiroz, J. Granados-Mortera, M. García-Cuevas, J. Morales-Gómez, A. González-Hernández, Successful pain management with epidural oxytocin, *CNS Neurosci. Ther.* 22 (2016) 532–534, <https://doi.org/10.1111/cns.12551>.
 - [9] G. Cirillo, D. de Luca, M. Papa, Calcium imaging of living astrocytes in the mouse spinal cord following sensory stimulation, *Neural Plast.* 2012 (2012) 1–7, <https://doi.org/10.1155/2012/425818>.
 - [10] D. Davalos, J.K. Lee, W.B. Smith, B. Brinkman, M.H. Ellisman, B. Zheng, K. Akassoglou, Stable in vivo imaging of densely populated glia, axons and blood vessels in the mouse spinal cord using two-photon microscopy, *J. Neurosci. Methods* 169 (2008) 1–7, <https://doi.org/10.1016/j.jneumeth.2007.11.011>.
 - [11] E.C. Emery, A.P. Luiz, S. Sikandar, R. Magnúsdóttir, X. Dong, J.N. Wood, In vivo characterization of distinct modality-specific subsets of somatosensory neurons using GCaMP, *Sci. Adv.* 2 (2016) e1600990, <https://doi.org/10.1126/sciadv.1600990>.
 - [12] M.J. Farrar, I.M. Bernstein, D.H. Schlafer, T.A. Cleland, J.R. Fetcho, C.B. Schaffer, Chronic in vivo imaging in the mouse spinal cord using an implanted chamber, *Nat. Methods* 9 (2012) 297–302, <https://doi.org/10.1038/nmeth.1856>.
 - [13] K.K. Fenrich, P. Weber, M. Hocine, M. Zalc, G. Rougon, F. Debarbieux, Long-term in vivo imaging of normal and pathological mouse spinal cord with subcellular resolution using implanted glass windows, *J. Physiol.* 590 (2012) 3665–3675, <https://doi.org/10.1113/jphysiol.2012.230532>.
 - [14] H.C. Johansson, F. Helmchen, Two-photon imaging of spinal cord cellular networks, *Exp. Neurol.* 242 (2013) 18–26, <https://doi.org/10.1016/j.expneurol.2012.07.014>.
 - [15] K.J. Sekiguchi, P. Shekhtmeyster, K. Merten, A. Arena, D. Cook, E. Hoffman, A. Ngo, A. Nimmerjahn, Imaging large-scale cellular activity in spinal cord of freely behaving mice, *Nat. Commun.* 7 (2016) 11450, <https://doi.org/10.1038/ncomms11450>.
 - [16] S. Vrontou, A.M. Wong, K.K. Rau, H.R. Koerber, D.J. Anderson, Genetic identification of C fibres that detect massage-like stroking of hairy skin in vivo, *Nature* 493 (2013) 669–673, <https://doi.org/10.1038/nature11810>.
 - [17] C. Luo, P.H. Seeburg, R. Sprengel, R. Kuner, Activity-dependent potentiation of calcium signals in spinal sensory networks in inflammatory pain states, *Pain* 140 (2008) 358–367, <https://doi.org/10.1016/j.pain.2008.09.008>.
 - [18] R. Ruscheweyh, J. Sandkühler, Long-range oscillatory Ca²⁺ waves in rat spinal dorsal horn, *Eur. J. Neurosci.* 22 (2005) 1967–1976, <https://doi.org/10.1111/j.1460-9568.2005.04393.x>.
 - [19] C.J. Woolf, M.W. Salter, Neuronal plasticity: increasing the gain in pain, *Science* 288 (2000) 1765–1768, <https://doi.org/10.1126/science.288.5472.1765>.
 - [20] M. Zimmermann, Ethical guidelines for investigations of experimental pain in conscious animals, *Pain* 16 (1983) 109–110, [https://doi.org/10.1016/0304-3959\(83\)90201-4](https://doi.org/10.1016/0304-3959(83)90201-4).
 - [21] E.A. Rendón-Ochoa, T. Hernández-Flores, V.H. Avilés-Rosas, V.A. Cáceres-Chávez, M. Duhne, A. Laville, D. Tapia, E. Galarraga, J. Bargas, Calcium currents in striatal fast-spiking interneurons: dopaminergic modulation of Ca V 1 channels, *BMC Neurosci.* 19 (2018) 42, <https://doi.org/10.1186/s12868-018-0441-0>.
 - [22] A. Aparicio-Juárez, M. Duhne, E. Lara-González, F. Ávila-Cascas, V. Calderón, E. Galarraga, J. Bargas, Cortical stimulation relieves parkinsonian pathological activity *in vitro*, *Eur. J. Neurosci.* 49 (2019) 834–848, <https://doi.org/10.1111/ejn.13806>.
 - [23] J. Pérez-Ortega, M. Duhne, E. Lara-González, V. Plata, D. Gasca, E. Galarraga, A. Hernández-Cruz, J. Bargas, Pathophysiological signatures of functional connectomics in parkinsonian and dyskinetic striatal microcircuits, *Neurobiol. Dis.* 91 (2016) 347–361, <https://doi.org/10.1016/j.nbd.2016.02.023>.
 - [24] L. Carrillo-Reid, F. Tecuapetla, D. Tapia, A. Hernández-Cruz, E. Galarraga, R. Drucker-Colín, J. Bargas, Encoding network states by striatal cell assemblies, *J. Neurophysiol.* 99 (2008) 1435–1450, <https://doi.org/10.1152/jn.01131.2007>.
 - [25] V. Plata, M. Duhne, J. Pérez-Ortega, R. Hernández-Martínez, P. Rueda-Orozco, E. Galarraga, R. Drucker-Colín, J. Bargas, Global actions of nicotine on the striatal microcircuit, *Front. Syst. Neurosci.* 7 (2013) 78, <https://doi.org/10.3389/fnsys.2013.00078>.
 - [26] O. Jáidar, L. Carrillo-Reid, A. Hernández, R. Drucker-Colín, J. Bargas, A. Hernández-Cruz, Dynamics of the Parkinsonian striatal microcircuit: entrainment into a dominant network state, *J. Neurosci.* 30 (2010) 11326–11336, <https://doi.org/10.1523/JNEUROSCI.1380-10.2010>.
 - [27] A.B. Salinas-Abarca, S.H. Avila-Rojas, P. Barragán-Iglesias, J.B. Pineda-Farías, V. Granados-Soto, Formalin injection produces long-lasting hypersensitivity with characteristics of neuropathic pain, *Eur. J. Pharmacol.* 797 (2017) 83–93, <https://doi.org/10.1016/j.ejphar.2017.01.018>.
 - [28] F.Y. Zhang, Y. Wan, Z.K. Zhang, A.R. Light, K.Y. Fu, Peripheral formalin injection induces long-lasting increases in cyclooxygenase 1 expression by microglia in the spinal cord, *J. Pain* 8 (2007) 110–117, <https://doi.org/10.1016/j.jpain.2006.06.006>.
 - [29] K.Y. Fu, A.R. Light, W. Maixner, Long-lasting inflammation and long-term hyperalgesia after subcutaneous formalin injection into the rat hindpaw, *J. Pain* 2 (2001) 2–11, <https://doi.org/10.1054/jpai.2001.9804>.
 - [30] E.W. Keefer, A. Gramowski, G.W. Gross, NMDA receptor-dependent periodic oscillations in cultured spinal cord networks, *J. Neurophysiol.* 86 (2001) 3030–3042, <https://doi.org/10.1152/jn.2001.86.6.3030>.
 - [31] J.D. Xie, S.R. Chen, H. Chen, W.A. Zeng, H.L. Pan, Presynaptic N-methyl-D-aspartate (NMDA) receptor activity is increased through protein kinase C in paclitaxel-induced neuropathic pain, *J. Biol. Chem.* 291 (2016) 19364–19373, <https://doi.org/10.1074/jbc.M116.732347>.
 - [32] Z. Hong-mei, Z. Li-jun, H.U. Xiao-dong, H.U. Neng-wei, Z. Tong, L.I.U. Xian-guo, H.-M. Zhang, L.-J. Zhou, X.-D. Hu, N.-W. Hu, T. Zhang, X.-G. Liu, Acute nerve injury induces long-term potentiation of C-fiber evoked field potentials in spinal dorsal horn of intact rat, *Sheng Li Xue Bao: [Acta Physiol. Sin.]* 56 (2004) 591–596 (Accessed 5 March 2020), <http://www.ncbi.nlm.nih.gov/pubmed/15497039>.
 - [33] H. Ikeda, J. Stark, H. Fischer, M. Wagner, R. Drdla, T. Jäger, J. Sandkühler, Synaptic amplifier of inflammatory pain in the spinal dorsal horn, *Science* 312 (2006) 1659–1662, <https://doi.org/10.1126/science.1127233>.
 - [34] N.A. Nelson, X. Wang, D. Cook, E.M. Carey, A. Nimmerjahn, Imaging spinal cord activity in behaving animals, *Exp. Neurol.* 320 (2019) 112974, <https://doi.org/10.1016/j.expneurol.2019.112974>.
 - [35] M. Condés-Lara, I.A.S. Maie, A.H. Dickenson, Oxytocin actions on afferent evoked spinal cord neuronal activities in neuropathic but not in normal rats, *Brain Res.* 1045 (2005) 124–133, <https://doi.org/10.1016/j.brainres.2005.03.020>.
 - [36] Y.H. Jo, M.E. Stoeckel, M.J. Freund-Mercier, R. Schlichter, Oxytocin modulates glutamatergic synaptic transmission between cultured neonatal spinal cord dorsal horn neurons, *J. Neurosci.* 18 (1998) 2377–2386 <http://www.ncbi.nlm.nih.gov/pubmed/9502799>.
 - [37] S. Caruso, C. Agnello, M.G. Campo, F. Nicoletti, Oxytocin reduces the activity of N-methyl — D — aspartate receptors in cultured neurons, *J. Endocrinol. Invest.* 16 (1993) 921–924, <https://doi.org/10.1007/BF03348959>.
 - [38] M. Mazzuca, M. Minlebaev, A. Shakirzyanova, R. Tyzio, G. Taccola, S. Janackova, S. Gataullina, Y. Ben-Ari, R. Giniatullin, R. Khazipov, Newborn analgesia mediated by oxytocin during delivery, *Front. Cell. Neurosci.* 5 (2011) 1–9, <https://doi.org/10.3389/fncel.2011.00003>.
 - [39] C.-Y. Jiang, T. Fujita, E. Kumamoto, Synaptic modulation and inward current produced by oxytocin in substantia gelatinosa neurons of adult rat spinal cord slices, *J. Neurophysiol.* 111 (2014) 991–1007, <https://doi.org/10.1152/jn.00609.2013>.
 - [40] G. Rojas-Piloni, M. López-Hidalgo, G. Martínez-Lorenzana, J. Rodríguez-Jiménez, M. Condés-Lara, GABA-mediated oxytocinergic inhibition in dorsal horn neurons by hypothalamic paraventricular nucleus stimulation, *Brain Res.* 1137 (2007) 69–77, <https://doi.org/10.1016/j.brainres.2006.12.045>.
 - [41] A. González-Hernández, G. Rojas-Piloni, M. Condés-Lara, Oxytocin and analgesia: Future trends, *Trends Pharmacol. Sci.* 35 (2014) 549–551, <https://doi.org/10.1016/j.tips.2014.09.004>.
 - [42] J. Qi, W.Y. Han, J.Y. Yang, L.H. Wang, Y.X. Dong, F. Wang, M. Song, C.F. Wu, Oxytocin regulates changes of extracellular glutamate and GABA levels induced by methamphetamine in the mouse brain, *Addict. Biol.* 17 (2012) 758–769, <https://doi.org/10.1111/j.1369-1600.2012.00439.x>.
 - [43] J.D. Breton, P. Veinante, S. Uhl-Bronner, A.M. Vergnano, M.J. Freund-Mercier, R. Schlichter, P. Poisbeau, Oxytocin-induced antinociception in the spinal cord is mediated by a subpopulation of glutamatergic neurons in lamina I-II which amplify GABAergic inhibition, *Mol. Pain* 4 (2008), <https://doi.org/10.1186/1744-8069-4-19> 1744–8069–4–19.
 - [44] W. Sun, Q. Zhou, X. Ba, X. Feng, X. Hu, X. Cheng, T. Liu, J. Guo, L. Xiao, J. Jiang, D. Xiong, Y. Hao, Z. Chen, C. Jiang, Oxytocin relieves neuropathic pain through GABA release and presynaptic TRPV1 inhibition in spinal cord, *Front. Mol. Neurosci.* 11 (2018) 248, <https://doi.org/10.3389/fnmol.2018.00248>.
 - [45] Y. Nerseyan, L. Demirkhanyan, D. Cabezas-Bratesco, V. Oakes, R. Kusuda, T. Dawson, X. Sun, C. Cao, A.M. Cohen, B. Chelluboina, K.K. Veeravalli, K. Zimmermann, C. Domene, S. Brauchi, E. Zakharian, Oxytocin modulates nociception as an agonist of pain-sensing TRPV1, *Cell Rep.* 21 (2017) 1681–1691, <https://doi.org/10.1016/j.celrep.2017.10.063>.